

Michaël Pons

Curriculum Vitae

(31 years old) – ✉ ponsm@gfz-potsdam.de

ABOUT ME

I am a geodynamicist specializing in numerical modelling to explore complex geodynamic and tectonic processes, from regional deformation to global mantle-lithosphere interactions. My research combines computational simulations with geological and geophysical data to address fundamental questions about Earth's dynamic evolution.

Please visit my personal website https://minerallo.github.io/website_MP/ for more details.

EDUCATION

2018 – 2022	PhD in Geophysics – (University of Potsdam - Germany)
2016 – 2017	Master 2/Postgraduate – Geodynamics - (University Rennes 1 - France)
2015 – 2016	Master 2/Postgraduate – Tectonics - Erasmus - (University Roma 3 - Italia)
2014 – 2015	Master 1/Graduate - Geodynamics - (University Orléans- France)
2013 – 2014	Licence /degree – Geology - Erasmus - (University College Cork - Ireland)
2011 – 2013	Licence /degree – Geology - (University Lille 1 - France)
2010 - 2011	Bacalaureat Scientific option Science of the Life and the Earth. - (France – Cannes)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher, GFZ, Germany

Jun 2022 - Present

Advisors: Prof. Stephan Sobolev (stephan.sobolev@gfz.de), Prof. Sascha Brune (sascha.brune@gfz.de)

- Built global-scale numerical models to investigate plate tectonics using the geodynamic code ASPECT
- Developed new features for ASPECT, including custom meshing and postprocessing modules for spherical geometries
- Integrated GPlately with the Geodynamic World Builder to define complex initial conditions for geodynamic simulations

PhD Researcher Universität Potsdam and GFZ, Germany

2018 – 2022

Advisors: Prof. Stephan Sobolev (stephan.sobolev@gfz.de), Prof. Frank Krüger (kruegerf@geo.uni-potsdam.de)

PhD thesis : “[The nature of tectonic shortening in the Central Andes](#)”

- Modeled Andean subduction dynamics with ASPECT to explore shortening variability
- Designed and implemented new postprocessing modules to track crustal shortening within the numerical models
- Developed a data-driven geodynamic model of the Pampean flat slab subduction to assess how slab geometry influences strain localization and foreland tectonic styles

MSc Researcher Université de Rennes I, France

2017

Advisors: Prof. Philippe Steer (philippe.steer@univ-rennes1.fr), Prof. Maxime Mouyen (maximemouyen@gmail.com), Prof. Laurent Longuevergne

MSc Thesis: *Earthquakes' impact on GRACE gravimetric signal at global scale*

- Developed a Matlab-based program to calculate coseismic displacements at the global scale
- Processed and filtered Level-2 GRACE data to isolate coseismic signals for comparison with numerical models

MSc Researcher Università Roma Tre, Italy

2016

Advisors: Prof. Claudio Faccenna (claudio.faccenna@uniroma3.it), Dr. Andrea Sembroni

MSc Thesis: *Topographic study of the Deccan region using geological and modeling approaches*

- Reconstructed pre-Deccan Traps topography ~65Ma ago
- Quantified isostatic flexural and dynamic topography to assess tectonic and mantle contributions

Research Intern Université d'Orléans, France

2015

Advisors: Prof. Yan Chen (yan.chen@univ-orleans.fr), Dr. Hongsheng Liu

Project: *Paleomagnetic study of the Qilitanling granitic massif (South China)*

- Applied AMS and paleomagnetic analyses (thermal demagnetization, Day plots) to the Qitianling pluton to investigate

its emplacement style

- Used paleomagnetic reconstructions using Paleomac to estimate the size of the Mongol-Okhotsk Ocean in the Late Jurassic

Research Intern *University College Cork, Ireland*

2014

Advisors: *Prof. Patrick. Meere* (p.meere@ucc.ie)

Project: *Finite strain analysis of conglomerates from the Dingle Peninsula*

- Quantified and mapped rock deformation using finite strain analysis techniques
- Analyzed conglomerates with varying grain sizes and distributions to assess strain patterns

TECHNICAL SKILLS

Programming languages :	MATLAB, C++, Python
Development tools :	Linux, git, bash scripting, High Performance Computing (SLURM)
Numerical Modelling :	ASPECT, GWB (worldbuilder), Gplates
Other software :	Paraview, GIS, Anisoft, Adobe illustrator

GRANT AND AWARDS

- Outstanding Student Poster, Ada Lovelace geodynamics Workshop, Hungary, Hévíz, 2022.
- Outstanding Student and PhD candidate Presentation (OSPP), EGU General Assembly 2022.
- Erasmus + traineeship grant, Roma Tre, 2016.
- Erasmus traineeship grant, Cork, 2014.

COMPUTATIONAL RESOURCES GRANTS

- HPC time on NHR@ZIB cluster, Germany
Project: Modelling support for monitoring Earth evolution through time (MSMEET) - [bbk00014](#)
Role: Co-investigator | Allocation: ~17 to 19 million CPU hours | 2024 - 2025 | ~€2 millions
- HPC time on NHR@ZIB cluster, Germany
Project: Plate boundary dynamics: Geodynamic modeling of the crust, lithosphere, and mantle – [bbp00039](#)
Role: Co-investigator | Allocation: ~10 to 19 million CPU hours | 2020 - 2024 | ~€2.9 millions
- HPC time on NHR@ZIB cluster, Germany
Project: Lower mantle dynamics: Modelling shear wave seismic anisotropy – [bbk00018](#)
Role: Co-investigator | Allocation: ~19 million CPU hours | 2024 - 2025 | ~€2 millions

TEACHING AND MENTORING

Teaching courses

- **Instructor**, *Online Summer School for Biogeodynamics*, “Tectonics numerical modelling” (2025)
- **Teaching Assistant**, *University of Potsdam (MSc level)*
 - Advanced Geodynamics: Practicals on numerical modeling (2025)
 - Neotectonics & Geodynamics: Sessions on deglaciation, dynamic topography, flexural isostasy, and fault reactivation (2023)

Mentoring for numerical modelling

I mentor researchers from PhD- to professor-level in numerical modelling, with a focus on using ASPECT for geodynamic simulations.

- Chaowei Lu, Melt weakening (2025 – present, with Xiaoping Yuan)
- Dr. Matias Barrionuevo, Andean deformation (2022)
- Dr. Miao Dong, plate tectonics and plumes dynamics (2024- present)

- Dr. Alexander Koptev, plume-rift interaction (2022 - present)
- Dr. Alessio Lavecchia, plumes-lithosphere delamination (2023 - present)
- Prof. Guido Gianni, subduction and lithosphere deformation (2023 - present)
- Tina Neumann, rift dynamics (2025 – present, with Sascha Brune)
- Guy Fisch, subduction dynamics (2024 – present, with Roi Granot and Sascha Brune)
- Madhusudan Sharma, subduction dynamics (2025 – present, with Ivone Jiménez Munt)
- Ying-ying Li, plume-subduction interaction (2024 – present, with Sascha Brune and Bernhard Steinberger)

SERVICES TO THE SCIENTIFIC COMMUNITY

Peer-reviews for scientific journals (2023-Present)

Geochemistry, Geophysics, Geosystems; Tectonics; Earth and Planetary Science Letters; Journal of Geophysical Research – Solid Earth

Session organization and chairing at GeoMOD 2025

Session “Tectonics and crustal deformation processes”

Session organization and chairing at EGU Gen. Assembly 2025

Session “Subduction Zone Dynamics: Global Perspectives with a focus on the western margin of South America”

Student poster/PICO award judge at EGU Gen. Assembly 2023-2025

Reviewed and provided feedback to graduate and undergraduate student talk and poster presentations

OUTREACH

Chief co-editor [EGU Geodynamics blog](#)

2022 - present

European Geoscience Union

- Manage team of 20 editors, columnists and illustrators, media managers
- Edit and write weekly blog posts related to Geodynamics and Academia
- Report the blog activity at EGU general assembly (>60k unique views/yr, over more than 120 countries)
- Written and edited over 20 blog posts since 2020. Here is a selection:

[The Nature of the Shortening in the Central Andes: Part 1](#) – Michael Pons (2023)

[Geodynamic Data-Driven Modelling: Bridging the Gap Between Observations and Numerical Models](#) - Constanza Rodriguez Picada (2023)

[A Journey to the Centre of Plate Tectonics and Mantle Convection](#) - João Duarte (2023)

[Modelling Magma Propagation in Three Dimensions](#) - Lorenzo Mantiloni (2023)

[How Has the Gibraltar Slab Influenced the Alboran Domain Deformation?](#) - Pedro J. Gea (2023)

[Reflections on the Geological and Geodynamic Evolution of the Southern Central Andes](#) - Matías Barrionuevo (2023)

[Why Do Seismic Images Vary Beneath Different Ocean Floors?](#) - Elodie Kendall (2022)

[How to Find Your Next Job in Earth Sciences?](#) - Michael Pons (2022)

[Rendering Tools for Geodynamics and Surface Modeling](#) - Michael Pons (2022)

Outreach representative for [EUROBIG](#)

2025 - present

European COST Action (Pan-EUROpean BioGeodynamics network)

- Developed, maintained and updated the EUROBIG website with project news and resources
- Developed outreach content to promote biogeodynamics research across disciplines
- Coordinated dissemination of activities and outcomes via web and social media channels

Science Communication & Visualization

2023

Berlin planetarium

- Created scientifically accurate visualizations of Earth's interior dynamics for public outreach and educational displays

LANGUAGES

CONTENT LINKS: Model examples

French (native) : Native

English : professional

⇒ [Global plate tectonics](#)

⇒ [Subduction dynamics](#)

Spanish : B1 intermediate
Italian : B1 beginner

⇒ [Subduction zone with surface processes](#)

PUBLICATIONS

- Li, Y., Steinberger, B., Brune, S., Le Breton, E. L., Glerum, A., & **Pons, M.** (submitted). Intraplate volcanism driven by slab–plume interaction: Numerical modeling and its application to the Eifel, Massif Central and Hainan volcanic areas. *Journal of Geophysical Research: Solid Earth*.
- Lei, Z., Xia, S., **Pons, M.**, & Brune, S. (submitted). Estimating rift setting’s potential to generate habitable bathymetry conditions for coral reefs: Spatio-temporal patterns through numerical modelling. *Journal of Geophysical Research: Solid Earth*.
- Li, K., Brune, S., Erdős, Z., Pons, M., & Glerum, A. (in preparation). The impact of orogenic inheritance on rifted margin formation.
- Alexander, A., Lavecchia, A., **Pons, M.**, Beekman, F., Brune, S., Cloetingh, S., Faccenna, C., Jolivet, L., Sobolev, S., & Tesauro, M. (in preparation). Numerical modelling of purely active (plume-produced) continental rifting and break-up. *Journal of Geophysical Research: Solid Earth*.
- Yuan, X., Brune, S., Braun, J., **Pons, M.**, Li, K., Li, Y., Shen, X., & He, X. (2024). *Coevolution of duplexing and crustal flow during Himalayan growth* (Version 1) [Preprint]. Research Square. Submitted to *Nature Geosciences*.
- Roy, P., Steinberger, B., Faccenda, M., & **Pons-Rallo, M. V.** (2025). Lowermost mantle flow at thermochemical piles constrained by shear wave anisotropy: Insights from combined geodynamic and mantle fabric simulations at global scale. *Authorea Preprints*. Submitted to *Geochemistry, Geophysics, Geosystems*
- Roy, P., Steinberger, B., Faccenda, M., & **Pons, M.** (2025). Modeling anisotropic signature of slab-induced mantle plumes from thermochemical piles in the lowermost mantle. *Geophysical Research Letters*, 52(10), e2024GL113299.
- Steinberger, B., Roy, P., **Pons, M.**, & Jopke, M. P. (2025). Why Are Plume Excess Temperatures Much Less Than the Temperature Drop Across the Lowermost-Mantle Thermal Boundary Layer?. *Journal of Geophysical Research: Solid Earth*, 130(4), e2024JB030111.
- Yuan, X., Li, Y., Brune, S., Li, K., **Pons, M.**, & Wolf, S. G. (2024). Coordination between deformation, precipitation, and erosion during orogenic growth. *Nature Communications*, 15(1), 10362.
- Pons, M.**, Rodriguez Picada, C., Sobolev, S. V., Scheck-Wenderoth, M., & Strecker, M. R. (2023). Localization of deformation in a non-collisional subduction orogen: The roles of dip geometry and plate strength on the evolution of the broken Andean foreland, Sierras Pampeanas, Argentina. *Tectonics*, 42(8), e2023TC007765.
- Pons, M.** (2023). *The Nature of the tectonic shortening in Central Andes* (Doctoral dissertation, Universität Potsdam).
- Pons, M.**, Sobolev, S. V., Liu, S., & Neuharth, D. (2022). Hindered trench migration due to slab steepening controls the formation of the Central Andes. *Journal of Geophysical Research: Solid Earth*, e2022JB025229.
- Rodriguez Picada, C., Scheck-Wenderoth, M., Bott, J., Gomez Dacal, M. Cacace, M.L., **Pons, M.**, Prezzi, C.B. & Strecker M., (accepted in Lithosphere) Controlling factors of the thermal field in an oceanic-continental subduction zone: the southern Central Andes
- Liu, S., Sobolev, S. V., Babeyko, A. Y., & **Pons, M.** (2022). Controls of the Foreland Deformation Pattern in the Orogen-Foreland Shortening System: Constraints from High-Resolution Geodynamic Models. *Tectonics*, 41(2), e2021TC007121.

CONFERENCE CONTRIBUTIONS

- Pons, M.**, Sobolev, S., & Jain, C. (2025). Plate boundaries dynamics between Earth and Venus. *Geomod 2025, Lisboa, Portugal, 8 September 2025*.
- Pons, M.**, Sobolev, S., & Jain, C. (2025). Subduction network connectivity: A comparison between Earth and Venus. *EGU General Assembly 2025, Vienna, Austria, 27 April – 2 May 2025*.
- Rodriguez Picada, C., **Pons, M.**, Scheck-Wenderoth, M., Cacace, M., Bott, J., & Strecker, M. (2025). Contributions of plate strength and dip geometry on the localization of deformation in Central Andes: A data-driven modelling approach. *EGU General Assembly 2025, Vienna, Austria, 27 April – 2 May 2025*.
- Yuan, X., Li, Y., Brune, S., Li, K., **Pons, M.**, & Wolf, S. (2025). How do deformation, orographic precipitation, and erosion coordinate during orogenic growth? *EGU General Assembly 2025, Vienna, Austria, 27 April – 2 May 2025*.
- Dong, M., **Pons, M.**, & Brune, S. (2025). Quantifying super-continent evolution through Wilson cycle phases at global scale. *EGU General Assembly 2025, Vienna, Austria, 27 April – 2 May 2025*.
- Roy, P., van Hunen, J., **Pons, M.**, & Chakraborty, A. (2025). Lithosphere-mantle interactions and weakening processes in early Earth: Implications on the onset of plate tectonics. *EGU General Assembly 2025, Vienna, Austria, 27 April – 2 May 2025*.
- Roy, P., Steinberger, B., Faccenda, M., Dannberg, J., & **Pons, M.** (2024). Shear wave radial anisotropy from slab-induced thermochemical plumes in the lower mantle. *Ada Lovelace workshop 2024*.
- Pons, M.**, Sobolev, S. V., Kendall, E., Jain, C., Roy, P., & Brune, S. (2024). Intensified plate tectonics by self-reorganisation of global subduction network post-Snowball Earth. *Ada Lovelace workshop 2024*.
- Li, K., Brune, S., Erdős, Z., **Pons, M.**, & Glerum, A. (2024). Quantifying the influence of orogenic inheritance on rifted margin formation through Wilson cycle modelling. *Ada Lovelace workshop 2024*.
- Pons, M.**, Sobolev, S., Jain, C., Kendall, E., & Davies, H. (2024, march). Global modeling of plate tectonics, from circum-supercontinental subduction to supercontinent dispersal. *Topic 3 GFZ Meeting*.

- **Pons, M.**, Sobolev, S. V., Piceda, C. R., Scheck-Wenderoth, M., Strecker, M., & Liu, S. (2024). Shortening mechanisms arising from Nazca plate subduction dynamics and its interaction with the South American plate. IPOC Meeting 2024.
- Li, K., Brune, S., Erdős, Z., Glerum, A., & **Pons, M.** : Numerical modelling of the Wilson Cycle: effects of orogenic inheritance on the formation of rifted continental margins, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024, EGU24-7743, <https://doi.org/10.5194/egusphere-egu24-7743>, 2024.
- Roy, P., Steinberger, B., Faccenda, M., Dannberg, J., & **Pons, M.** : Exploring the Development of Shear Wave Radial Anisotropy in the Lower Mantle due to Slab-induced Plume Generation from LLSVPs, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024, EGU24-13585, <https://doi.org/10.5194/egusphere-egu24-13585>, 2024.
- Steinberger, B., Roy, P & **Pons, M.** : Why are plume excess temperatures much less than the temperature drop across the core-mantle boundary?, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024, EGU24-11566, <https://doi.org/10.5194/egusphere-egu24-11566>, 2024.
- **Pons, M.**, Sobolev, S. V., Jain, C., & Kendall, E. (2024). From plumes to subduction network formation and supercontinent break-up (No. EGU24-10967). Copernicus Meetings.
- Sobolev, S., Jain, C., & **Pons, M.** (2024). Why does Earth have plate tectonics and what was before? (No. EGU24-12766). Copernicus Meetings.
- **Pons, M.**, Rodriguez Piceda, C., Sobolev, S. V., Scheck-Wenderoth, M., & Strecker, M. R. (2023, May). Understanding the role of structural inheritance and flat slab geometry in Central Andes. In *EGU General Assembly Conference Abstracts* (pp. EGU-8492).
- **Pons, M.**, Sobolev, S., Kendall, E., & Davies, H. (2023). From regional to global scale modeling of plate tectonics evolution and its interplay with surface processes. German-Swiss Geodynamics Workshop 2023.
- **Pons, M.**, Sobolev, S.V., Piceda, C. R., Sibiao, L., Neuharth, D., Scheck-Wenderoth, M., & Strecker, M. (2022, August). Plate interaction, subduction dynamics the role of the flat-slab subduction in the Central Andes. Ada Lovelace workshop 2022.
- **Pons, M.**, Sobolev, S.V., Sibiao, L., & Neuharth, D. (2022, May). Variability of the shortening rate in Central Andes controlled by subduction dynamics and interaction between slab and overriding plate. EGU conference 2022.
- **Pons, M.**, & Sobolev, S.V., (2021, September). Control of subduction dynamics on shortening magnitude in the Central Andes: a thermomechanical modeling approach. GEOMOD conference 2021.
- **Pons, M.**, & Sobolev, S.V., (2021, August). Interplay between the shortening magnitude and subduction dynamics in the Central Andes. Geodynamics WS 2021.
- Piceda, C. R., Scheck-Wenderoth, M., Bott, J., Dacal, M. L. G., **Pons, M.**, Prezzi, C., & Strecker, M. (2021). Unravelling the thermal state of the southern Central Andes and its controlling factors (No. EGU21-5214). Copernicus Meetings.
- **Pons, M.**, & Sobolev, S., (2020, May). The nature of the North-South change of the magnitude of tectonic shortening in Central Andes at Altiplano-Puna latitudes: a thermomechanical modeling approach. In EGU General Assembly Conference Abstracts (p. 8177).
- **Pons, M.**, & Sobolev, S.V., (2019). The nature of the North-South change of the magnitude of tectonic shortening in Central Andes at Altiplano-Puna latitudes: a thermomechanical modeling approach. 25th Latin-American Colloquium (LAC), Hamburg, Germany. Program and Abstracts (p. 61).
- **Pons, M.**, Sobolev, S.V., Liu, S., Glerum, A Rodriguez Piceda, C., (2019). The nature of the North-South change of the magnitude of tectonic shortening in Central Andes at Altiplano-Puna latitudes: a thermomechanical modeling approach. 5th International Young Earth Scientists Congress (YES). Berlin. Book of abstracts (p.49).
- **Pons, M.**, Sobolev, S.V., Liu, S., Glerum, A Rodriguez Piceda, C., (2019). The nature of the North-South change of the magnitude of tectonic shortening in Central Andes at Altiplano-Puna latitudes: a thermomechanical modeling approach. Topoeurope, Granada.

INVITED TALKS

- **Pons, M** (2025). Plate boundaries dynamics between Earth and Venus, GeoMOD 2025, Lisboa, Portugal.
- **Pons, M** (2025). Understanding the Earth, from the Andes to plate tectonics, COMTEC Argentinian geological survey, seminar series.
- **Pons, M.** (2024, may). From regional subduction (e.g., Andean-type) to global plate tectonics. Plymouth University, UK.
- **Pons, M.** (2024, March). Invited talk: From regional subduction (e.g., Andean-type) to global plate tectonics modelling. Aspect User Meeting 2024.
- **Pons, M.** (2023, july) ASPECT – FastScape coupling and then? ASPECT Hackaton 2023. Oregon, US.
- **Pons, M.** (2023, January). Invited talk: The nature of the tectonic shortening in the Central Andes: A geodynamic perspective. Freie Universität Berlin.

PUBLICATIONS NEAR SUBMISSION

- Koptev, A., Cloetingh, S., **Pons, M.**, Sobolev, S., Lavecchia, A., Tesauro, M., Brune, S., Faccenna, C., Jolivet, L., & Beekman, F. (in preparation). A geodynamic multicomponent ecosystem for continental rifting in East Africa.